7.3. Morphisms of functors

One of the important features of functors between categories which distinguishes them from usual maps or functions is that the functors between two given categories themselves form a category; i.e., one can define a nontrivial notion of a morphism between two functors.

Definition 7.3.1. Let $\mathcal{C}, \mathcal{D}$ be categories and let $F, G : \mathcal{C} \to \mathcal{D}$ be functors between them. A morphism $a : F \to G$ (also called a **natural transformation** or a **functorial morphism**) is a collection of morphisms $a_X : F(X) \to G(X)$ labeled by the objects X of $\mathcal{C}$, which is functorial in X; i.e., for any morphism $f : X \to Y$ (for $X, Y \in \mathcal{C}$) one has $a_Y \circ F(f) = G(f) \circ a_X$.

A morphism $a : F \to G$ is an isomorphism if there is another morphism $a^{-1} : G \to F$ such that $a \circ a^{-1}$ and $a^{-1} \circ a$ are the identities. The set of morphisms from F to G is denoted by $\mathrm{Hom}(F, G)$.

Example 7.3.2. 1. Let $\mathbf{FVect}_k$ be the category of finite dimensional vector spaces over k. Then the functors id and $**$ on this category are isomorphic. The isomorphism is defined by the standard maps $a_V : V \to V^{**}$ given by $a_V(u)(f) = f(u)$, $u \in V$, $f \in V^*$. But these two functors are not isomorphic on the category of all vector spaces $\mathbf{Vect}_k$, since for an infinite dimensional vector space V, V is not isomorphic to V^{**}.

2. Let $\mathbf{FVect}'_k$ be the category of finite dimensional k-vector spaces, where the morphisms are the isomorphisms. We have a functor F from this category to itself sending any space V to V^* and any morphism a to $(a^*)^{-1}$. This functor satisfies the property that V is isomorphic to $F(V)$ for any V, but it is not isomorphic to the identity functor. This is because the isomorphism $V \mapsto F(V) = V^*$ cannot be chosen to be compatible with the action of $GL(V)$, as V is not isomorphic to V^* as a representation of $GL(V)$.

3. Let A be an algebra over a field k, and let $F : A - \mathbf{mod} \to \mathbf{Vect}_k$ be the forgetful functor. Then as follows from Problem 2.3.17, $\mathrm{End} F = \mathrm{Hom}(F, F) = A$.

4. The set of endomorphisms of the identity functor on the category $A - \mathbf{mod}$ is the center of A (check it!).